



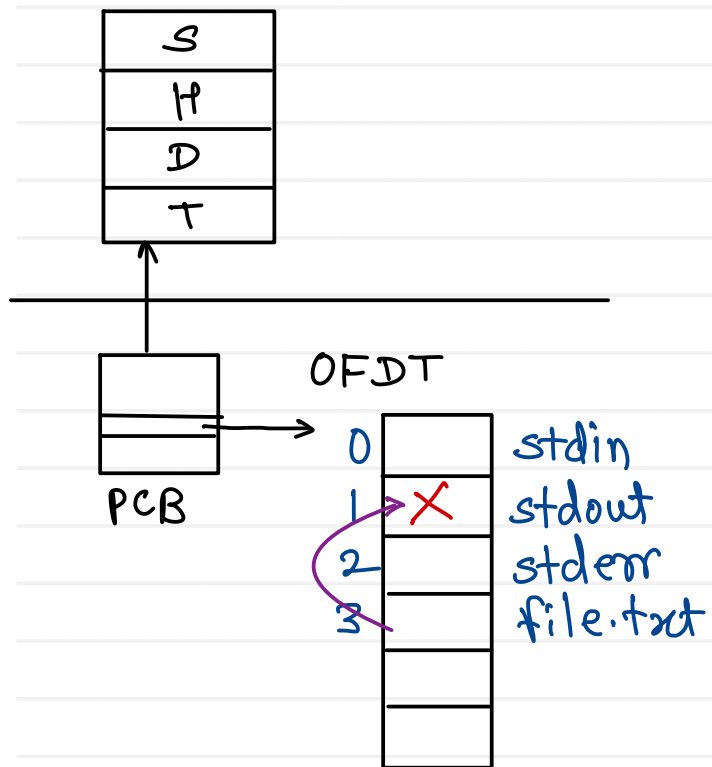
**Sunbeam Institute of Information Technology
Pune and Karad**

Module - Concepts of Operating Systems and Administration

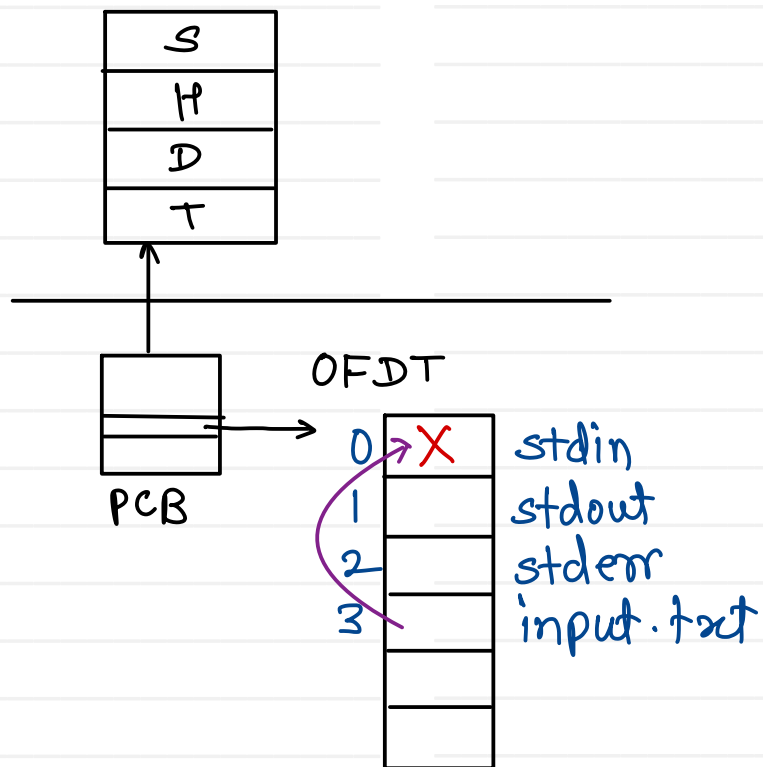
Trainer - Devendra Dhande

Email – devendra.dhande@sunbeaminfo.com

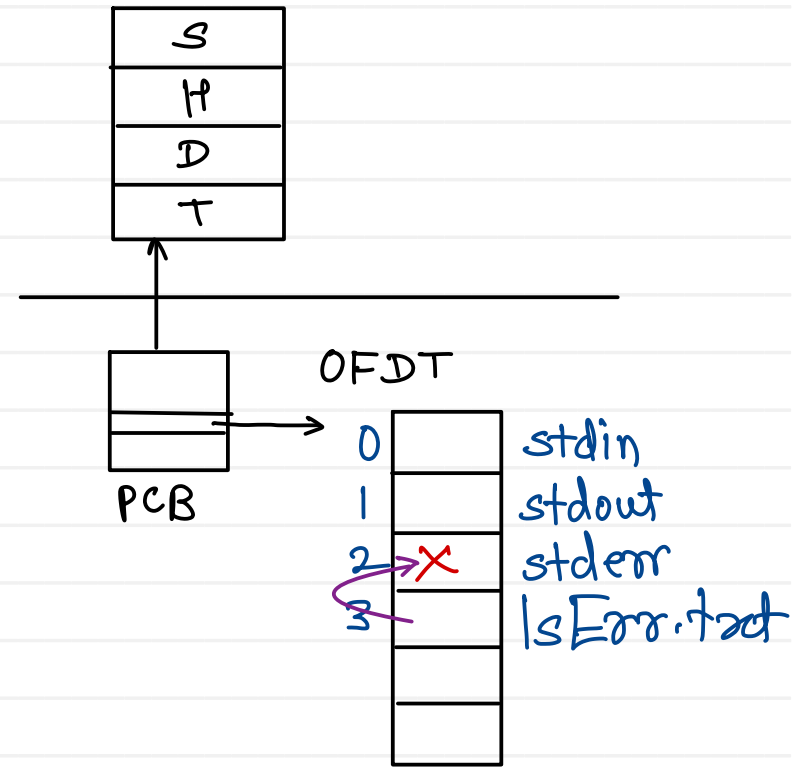
`cat > file.txt`



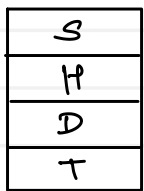
`cat < input.txt`



`ls -W 2> lsErr.txt`



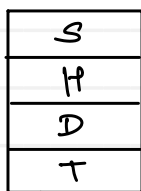
writer
process



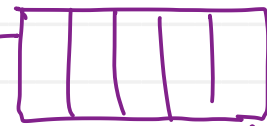
OFDT



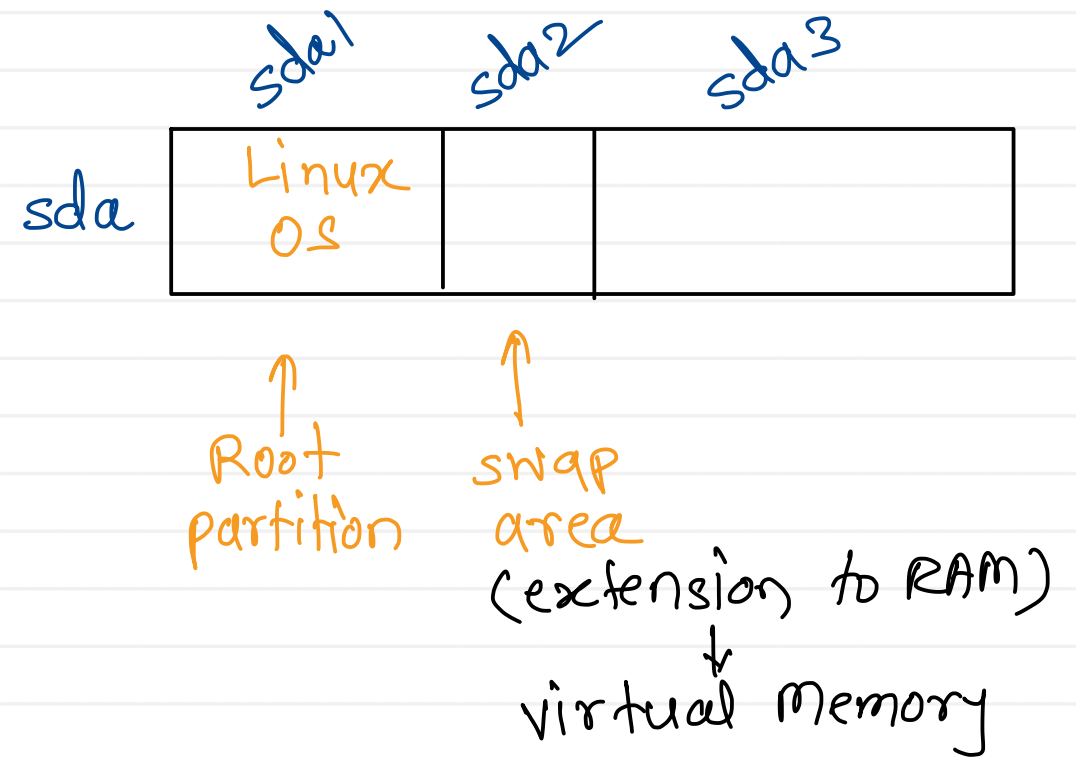
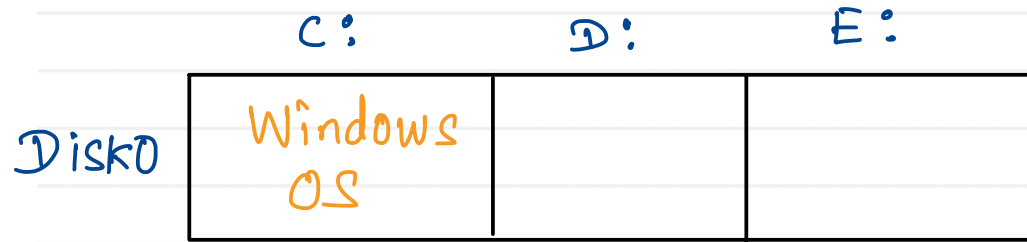
reader
process



OFDT



circular buffer

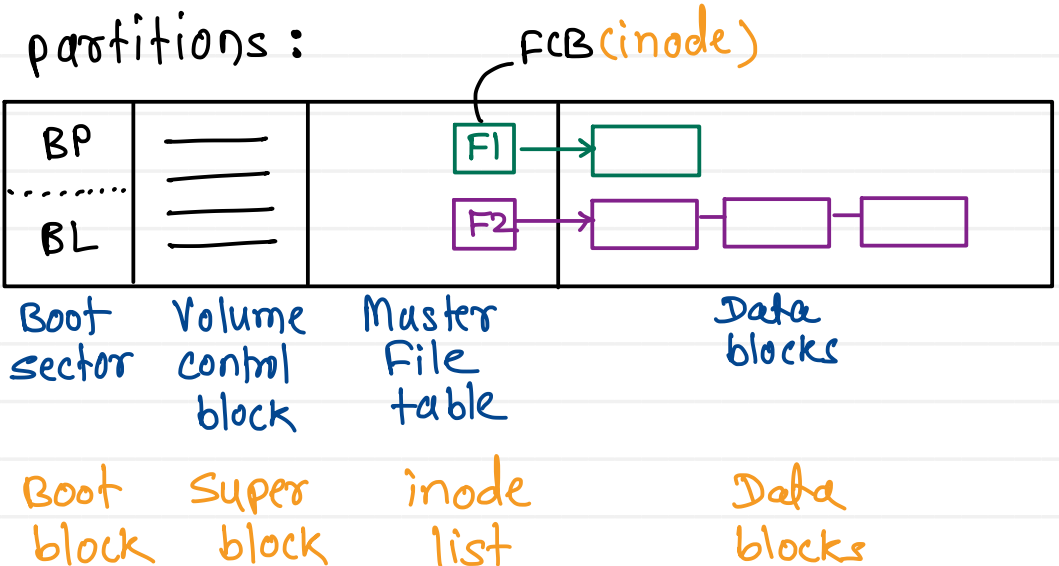


File : collection of data
 File : data + metadata
 (actual content of file)
 (information about file)

(Data block)
 (File Control Block)
 (FCB)

- file name
- owner / group
- timestamps
create / modify / access
- permissions
r-read, w-write, x-execute
levels - user / group / others
- size
- type
- link count
- info about data blocks

File system : organizing files on partitions



File - types, permissions, groups

type → - rw-rw-r-- 1 sunbeam sunbeam 86 Sep 23 04:33 move.txt
 permissions owner/user group size (in bytes) time stamp name

Types of file :

1. Regular file (-)
2. Directory file (d)
3. Link file (l)
4. Pipe file (p)
5. Socket file (s)
6. Char special file (c)
7. Block special file (b)

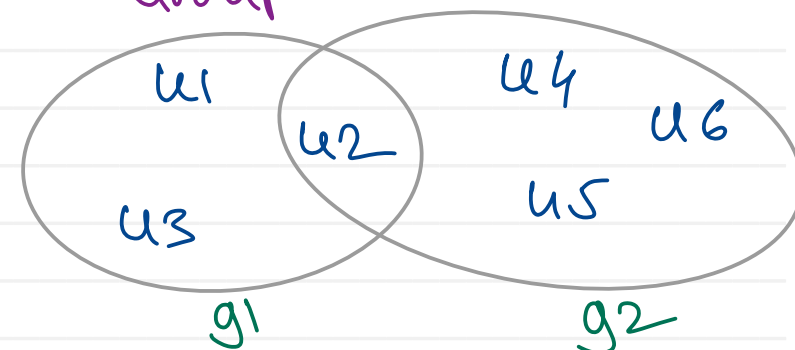
Permissions :

r - read
 w - write
 x - execute

levels :

1. user/owner
2. group
3. others

Group :



file name : move.txt
 owner : u1 (rw-)
 group : g1(u2, u3) (rw-)
 others : u4, u5, u6 (r--)

r = read, w = write, x = execute

$+$: add

$-$: remove

$\text{chmod } \pm r/w/x \text{ filename}$

e.g. $\text{chmod } -r \text{ file.txt}$

$\text{chmod } +x \text{ file.txt}$

u - user

g - group

o - others

$\text{chmod } u/g/o \pm r/w/x \text{ filename}$

e.g. $\text{chmod } u+x \text{ file.txt}$

$\text{chmod } g-rw \text{ file.txt}$

0 - 000

7 - 111

Human readable format : $rwX --- r--$

Octal format : 704

rwX	$---$	$r--$
111	000	100
$\hline 7$	$\hline 0$	$\hline 4$

Existing perms : $rwX --- r--$

New perms : $rw-rw-r--$

rwX	$---$	$r--$
111	000	100
110	110	100
$\hline 6$	$\hline 6$	$\hline 4$

$\Rightarrow (664)_8$

$\text{chmod } (mode)_8 \text{ filename}$

e.g. $\text{chmod } 664 \text{ file.txt}$